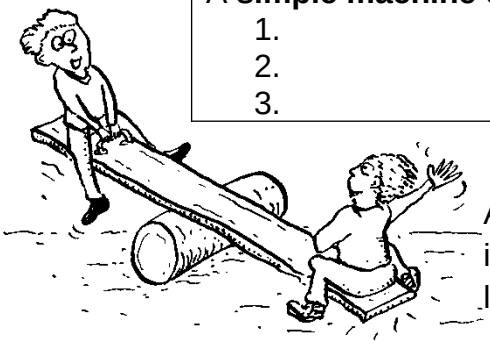
SIMPLE MACHINES: Levers

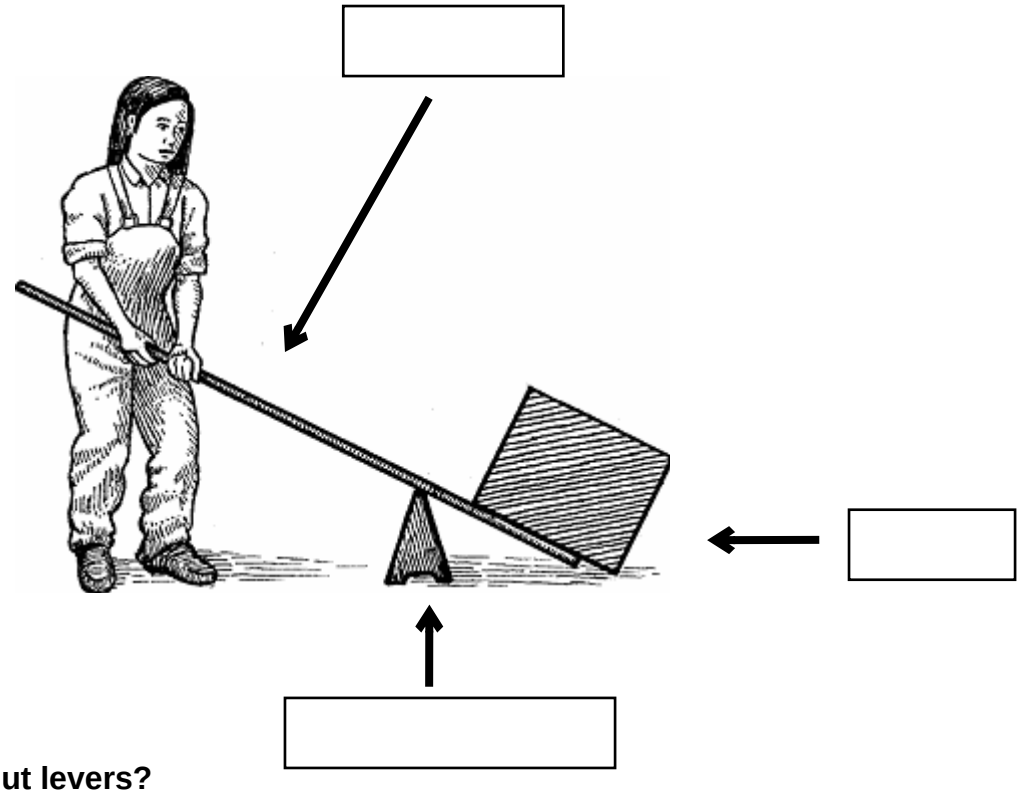
Name:	Date:	Teacher:



A **simple machine** can make a task easier in three ways:

1.
2.
3.

A **lever** is a bar such as stick or metal rod and a **fulcrum** which it pivots on. Most levers decrease the effort needed to move a load.



What do you know about levers?

Statement	True/False/Not sure
1. The point that a lever turns about is called the fulcrum.	
2. The fulcrum is represented in drawings as a square.	
3. A force applied to a lever is called a downward effort.	
4. The object that is being shifted is called the load.	
5. The direction in which the effort or load on a lever is acting is shown in arrows	
6. The distance between the fulcrum and the load is called the load arm.	
7. The force advantage would apply when a large effort moves a small load.	
8. Levers can give a force advantage or a distance advantage	

ACTIVITIES

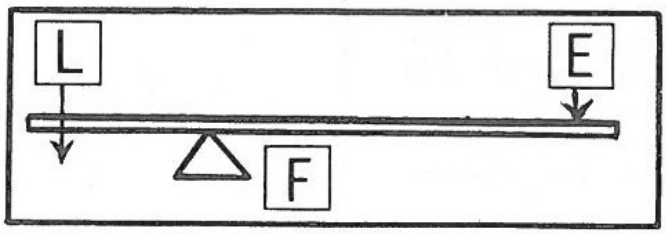
LSS Measurement Activity 26 Levers are useful

Equipment :

lever fulcrum load (slotted masses)
spring balance 2.5N, 5N, 50N paperclip

Type One Lever: a type one lever has the fulcrum positioned between the effort and the load.





L= LOAD F = FULCRUM E= EFFORT

Part 1 Method:

1. Set up the lever so that the **fulcrum is 20cm** from one end.
2. Hang a **300g load** from the end closest to the fulcrum [**0cm].**
3. Using a paperclip hang the 2.5N **spring balance to the 100cm mark** and pull downwards on the hooked end until the lever is level. Record the amount of effort (force) it takes to hold the lever horizontally.
4. Repeat for the next 2 loads.

Results:

LOAD (g)	EFFORT (N)
300	
400	
500	

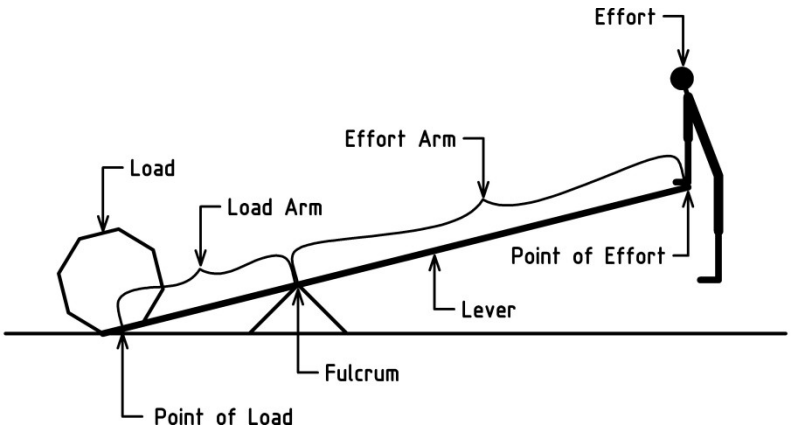
Discussion:

How did the effort change with the load?

Part 2 Method:

LSS Measurement Activity 26 Levers are useful

1. Set up the lever so that the **fulcrum is 60cm** from one end.
2. Hang a **300g load** from the end closest to the fulcrum [0cm].
3. Using a paperclip hook the 5N **spring balance to the 100cm mark** and pull downwards on the hooked end until the lever is level. Record the amount of effort (force) it takes to hold the lever horizontally.
4. Repeat, **changing the fulcrum** each time but keeping the **load the same**.



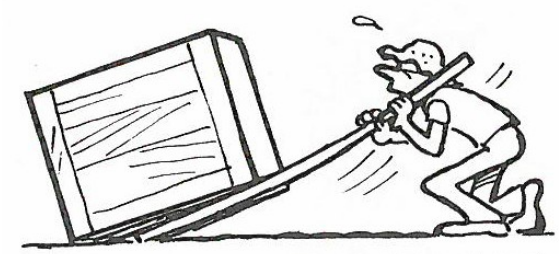
Results:

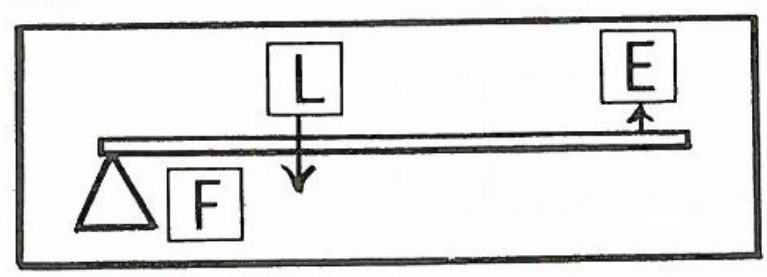
Fulcrum position (cm)	Effort arm length (cm)	Effort (N)
60		
70		
80		
90		

Discussion:

What happens to the effort required as the load arm increases in size?

Type Two Lever: a type two lever has the load positioned between the effort and the fulcrum.





L= LOAD F = FULCRUM E= EFFORT

Part 3 Method:

1. Set up the lever with the **fulcrum at 0cm**.
2. Hang a **300g load at 50cm**.
3. Using a paperclip hang the **2.5N spring balance at the 100cm** mark and **lift** upwards using the hooked end.
4. Record the effort taken to lift the lever.
5. Repeat, increasing the load each time.

Results:

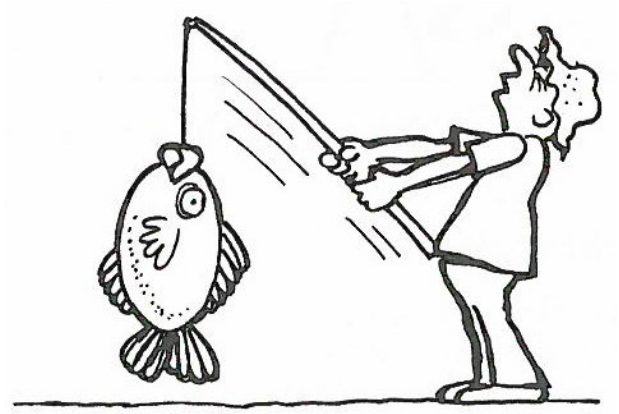
Load (N)	Effort (N)
300	
400	
500	

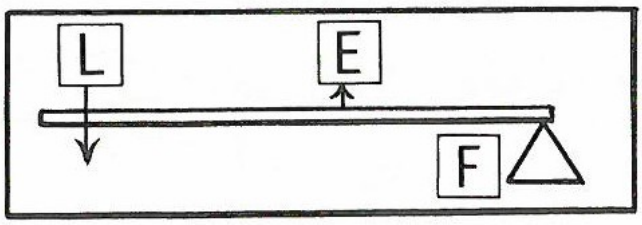
Discussion:

How did the effort change with the load?

Is there a force advantage when using a type two lever?

Type Three Lever: a type three lever has the effort between the load and fulcrum position.





Part 4 Method:

LSS Measurement Activity 26 Levers are useful

1. Set up the lever with the **fulcrum at 50cm**.
2. Hang **300g on the 100cm mark**.
3. Using a paperclip hang the 50N **spring balance at the 75cm mark** and **LIFT upwards using the hooked end**
4. Record the effort needed to hold the lever horizontal.
5. Repeat, increasing the load each time.

Results:

Load (g)	Effort (N)
300	
400	
500	

Discussion:

What happened to the effort as the load increased?

Is there a force advantage using a type three lever? Explain.

The lever as a simple machine

The pivot point (a fixed point) is called the **fulcrum**, this is shown as a triangle. When you apply a force to the lever this is called the **effort** and an arrow shows if you are pushing up or down. The object you are trying to move is called the **load**, an arrow shows where it is acting.

The **load arm** is the distance from the load to the fulcrum.

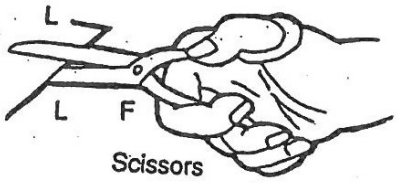
The **effort arm** is the distance from the effort to the fulcrum.

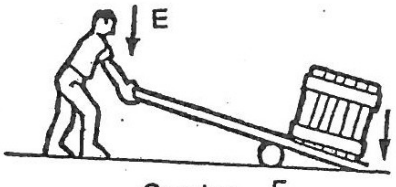
A lever can be used to gain a **force advantage** OR a **distance advantage**.

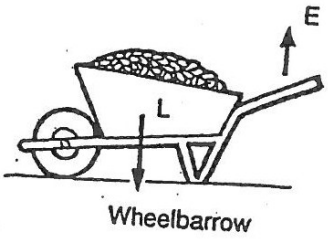
A **force advantage** is gained when the effort force used is **less** than the load force that is moved.

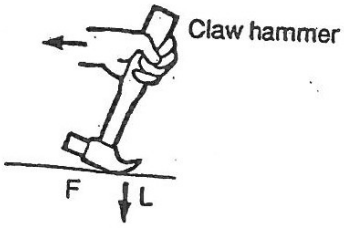
A **distance advantage** is gained when the effort force is moved through a **shorter** distance than the load force.

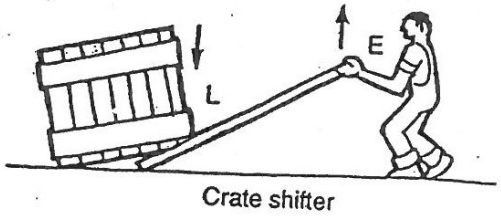
Classify each of the following as 1st, 2nd or 3rd type levers.

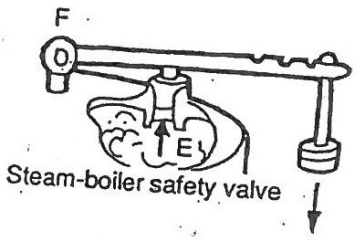

L
L F
Scissors

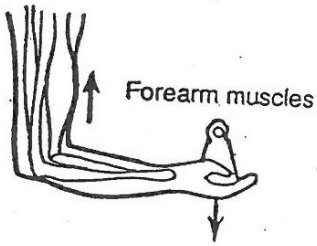

E
Crowbar F

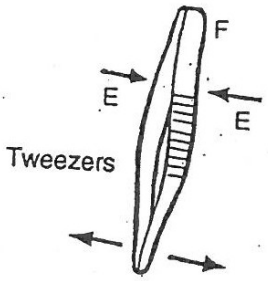

E
L
Wheelbarrow

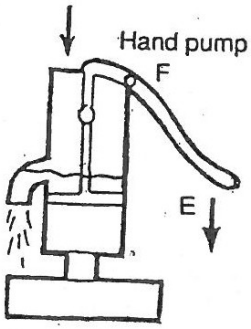

Claw hammer
F
L

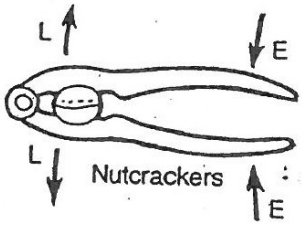

E
L
Crate shifter

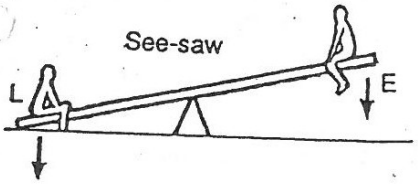

F
E
Steam-boiler safety valve

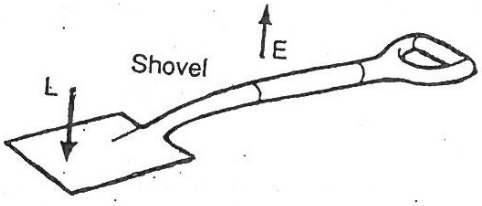

Forearm muscles

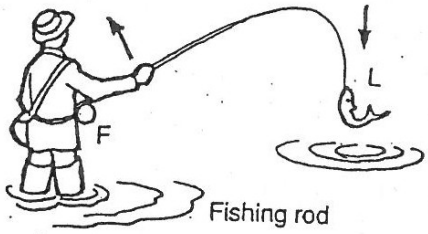

F
E
E
Tweezers

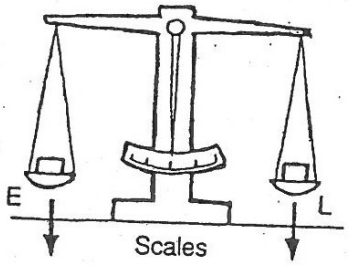

Hand pump
F
E

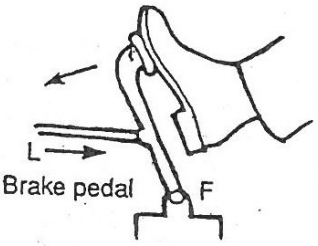

L
E
L
Nutcrackers
E


See-saw
L
E


Shovel
E
L


L
F
Fishing rod


E
L
Scales


L
Brake pedal
F

Identify the type of lever they are (1st, 2nd or 3rd). Locate and label the levers with effort, load and fulcrum for each example.



